

PRESENTATION SCIENTIFIC COMPUTING

LINMA2710

D. Guerand

CONTENTS

1. Part 1 : Basic matrix operations and SIMD
2. Part 2 : OpenMP
3. Part 3 : Distributed matrix operations (MPI)
4. Part 4 : GPU Matrix Operations (OpenCL)

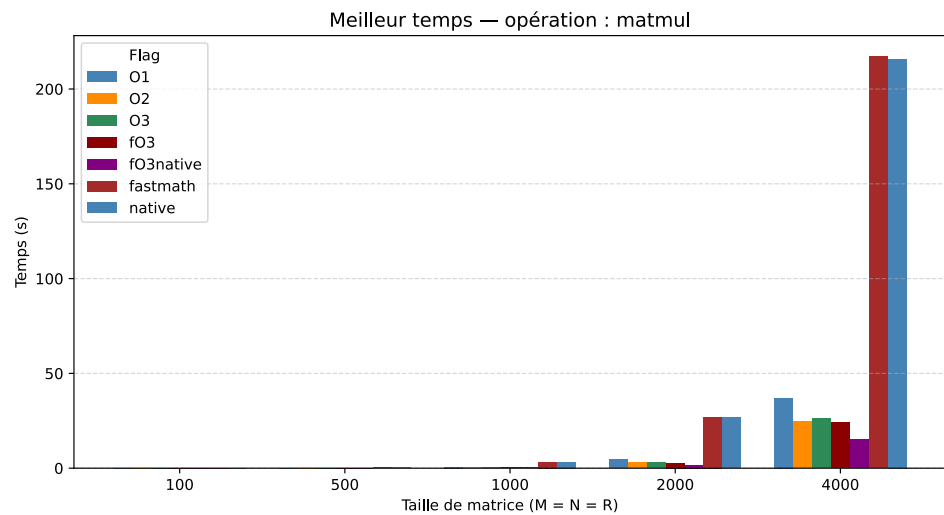
PART 1 : BASIC MATRIX OPERATIONS AND SIMD

- Matrices stored in **row-major** order → transposing $\mathbf{B}^T$ before multiply avoids column-stride cache misses
- **Tiling** splits the computation into small blocks that fit in L1/L2 cache (detailed in Part 2)
- Compiler flags: 01, 02, 03 -march=native enable auto-vectorisation (SIMD)

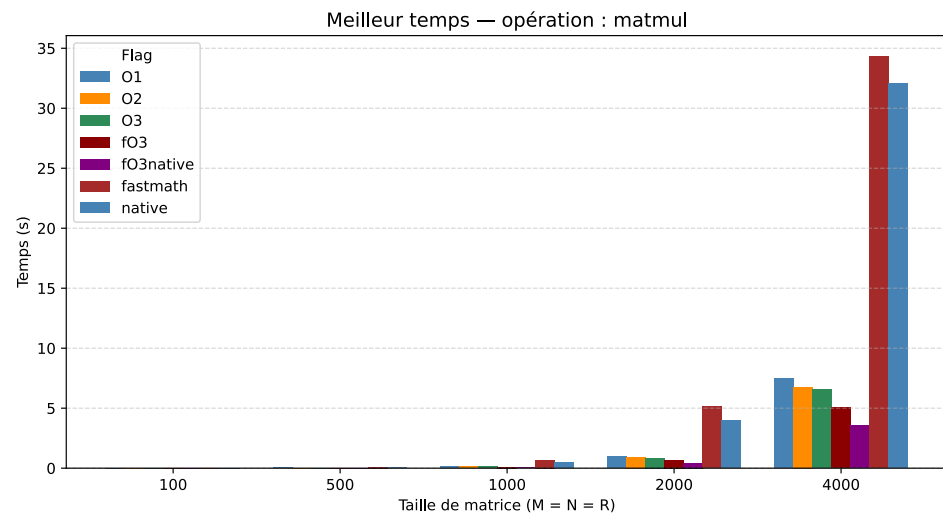
Key takeaway

Transpose + tiling together bring the matmul close to the **roofline bound**.

Server (avec tilling)



My PC

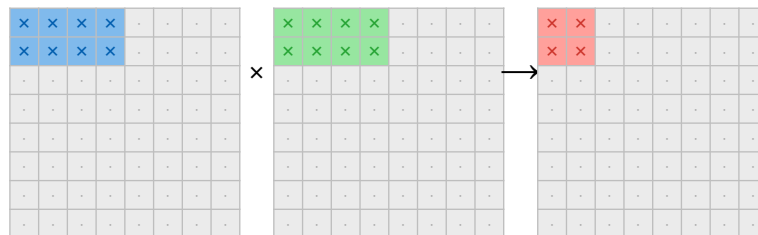


PART 2 : OPENMP

Matrice 8×8, proportionnel à TILE 64/64/128

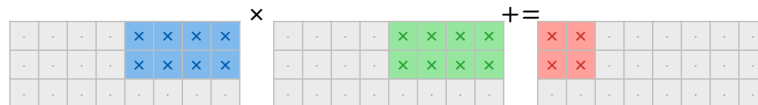
$ii=0, jj=0, kk=0$ — premier tile de C

A B^T C



$ii=0, jj=0, kk=4$ — accumulation (même case de C)

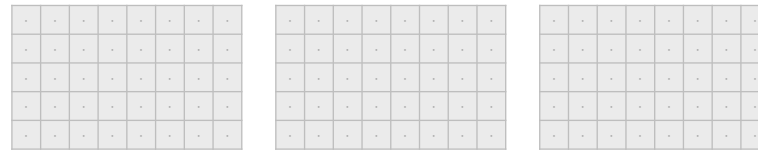
A B^T C



- Découpe le calcul en **blocs (tiles)** qui tiennent en cache L2
- Le tile de A reste en cache pendant toute la boucle jj
- Relu C/TILE_J fois **depuis le cache**, pas depuis la RAM
- ↑ arithmetic intensity → approche la limite roofline

Tailles de tiles choisies

- **Mon PC** : $\text{TILE_I} = \text{TILE_J} = \text{TILE_K} = 64$
- **Serveur** : $\text{TILE_I} = \text{TILE_J} = 64,$
 $\text{TILE_K} = 128$

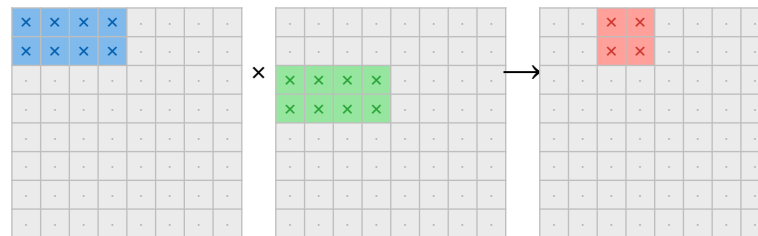


ii=0, jj=2 — tile suivant de C

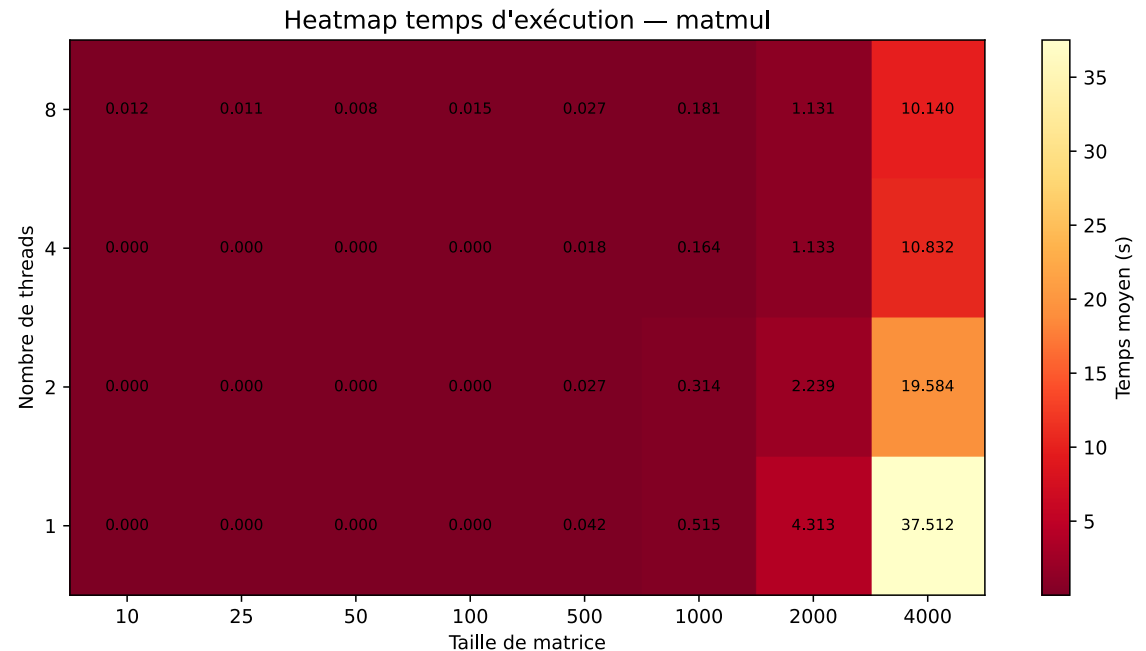
A

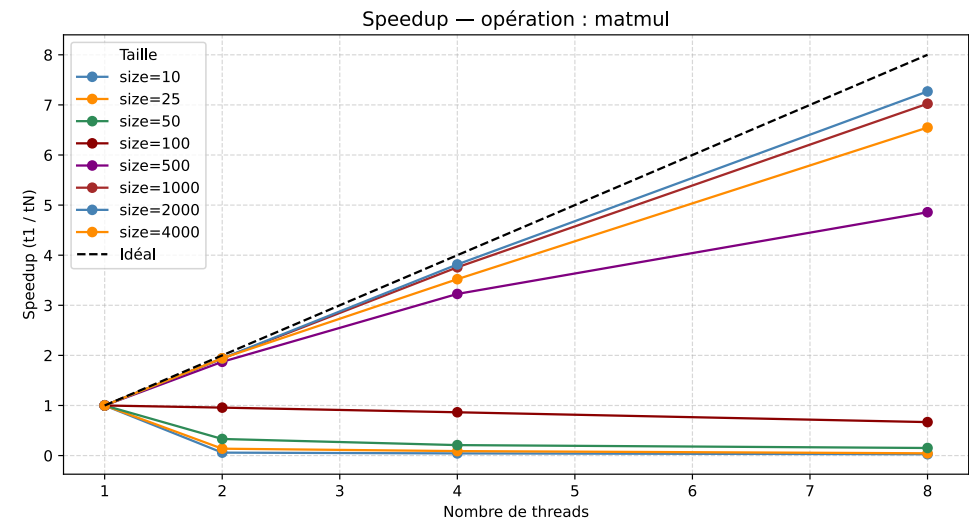
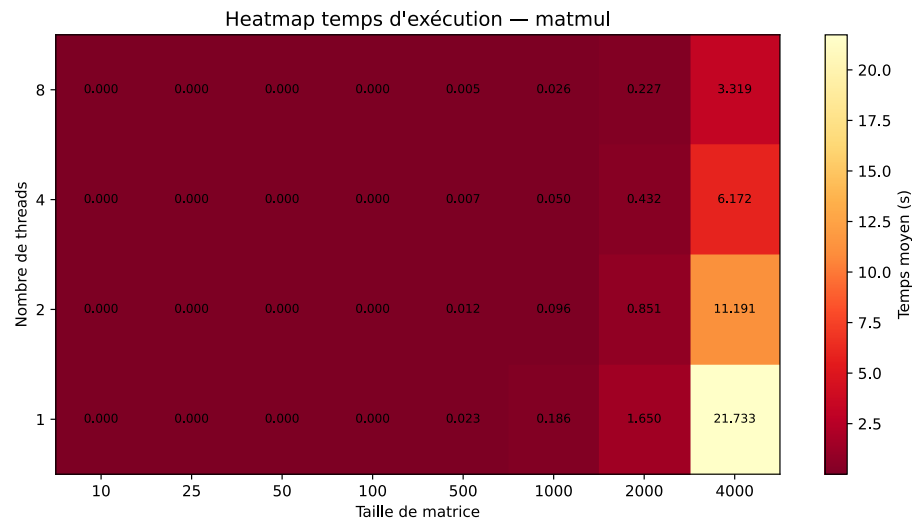
B^T

C



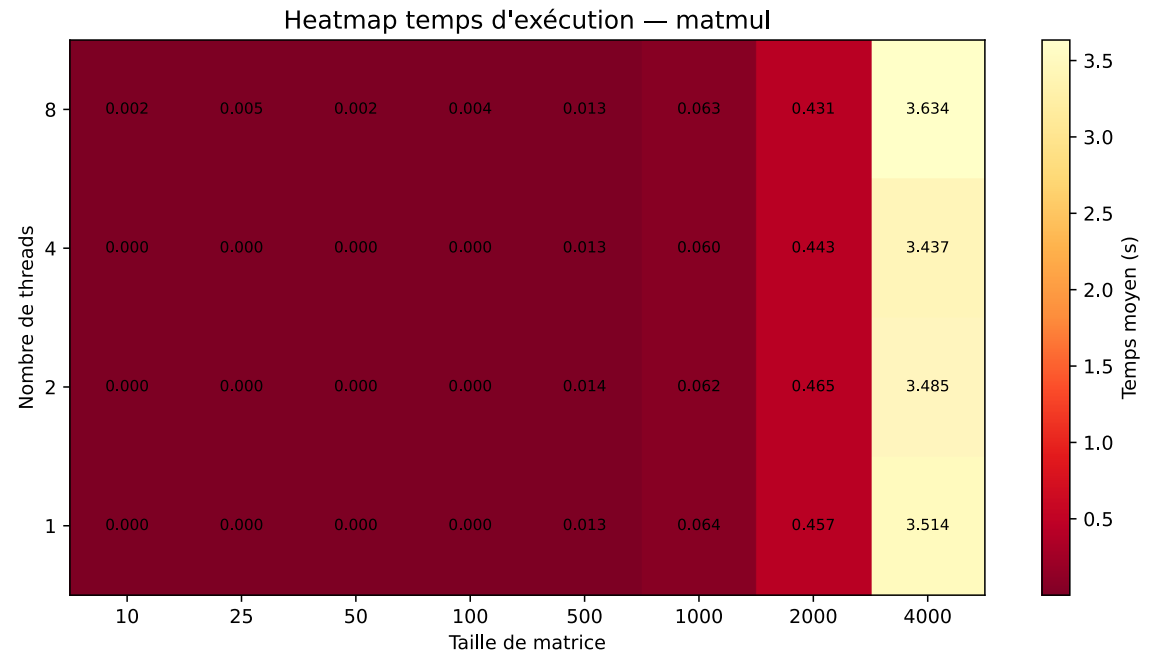
- Pour les **petites matrices** : le temps de création des threads domine
- Saturation entre 4 et 8 threads (nb de cœurs physiques)
- Speedup sous-linéaire dû aux contentions mémoire

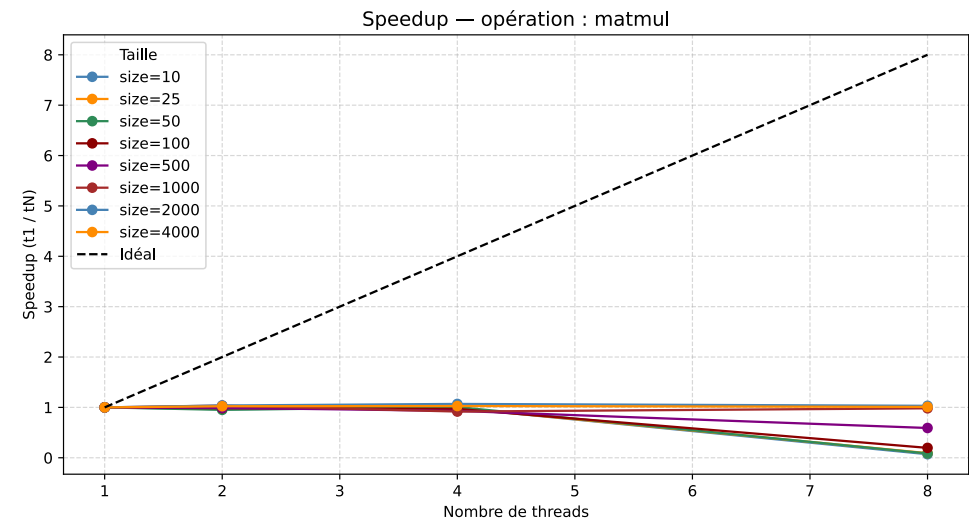
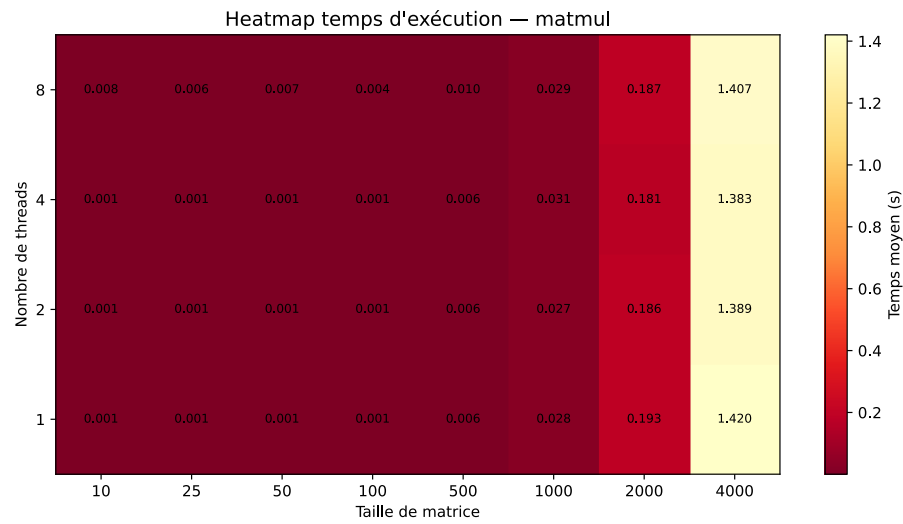




- Modèle roofline : limitation de la **bande passante mémoire**
- Le tilling sature la BW → peu de gain avec plus de threads

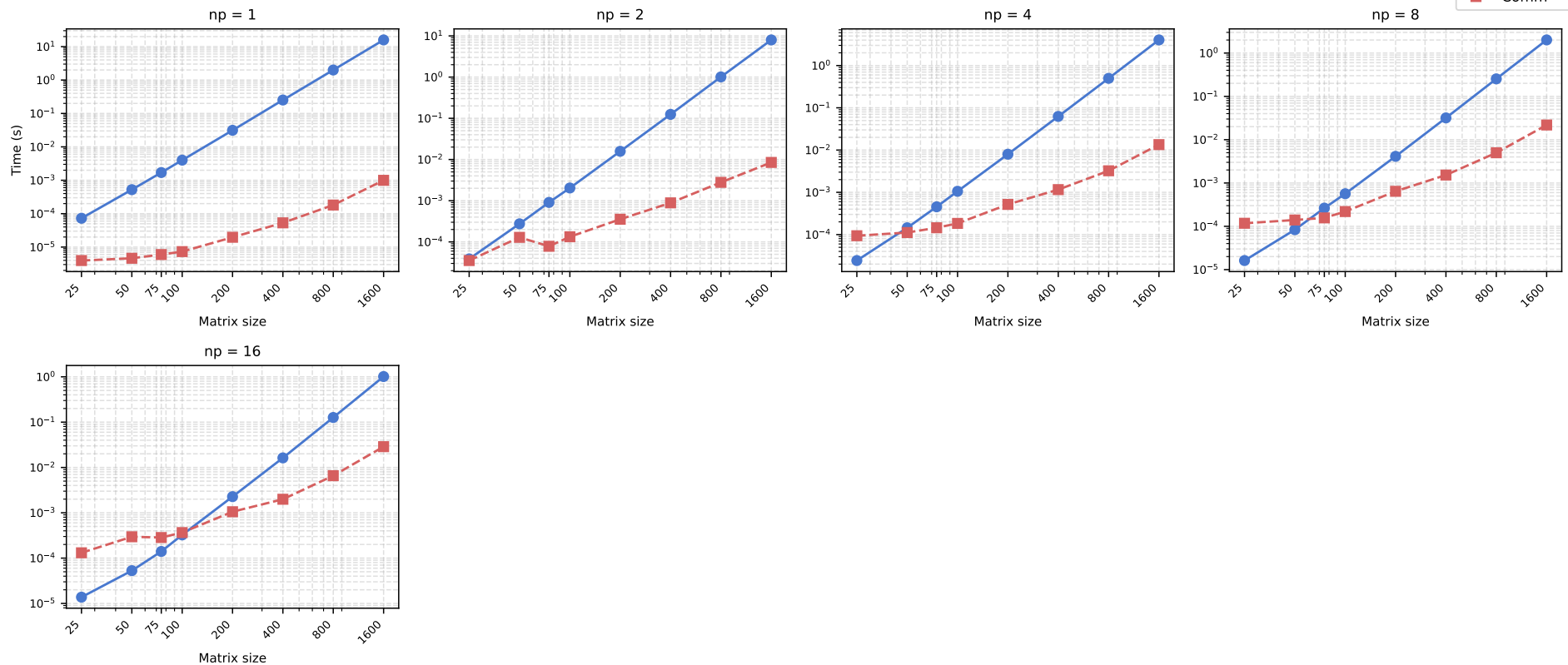
Le bottleneck passe de **compute** à **mémoire**.



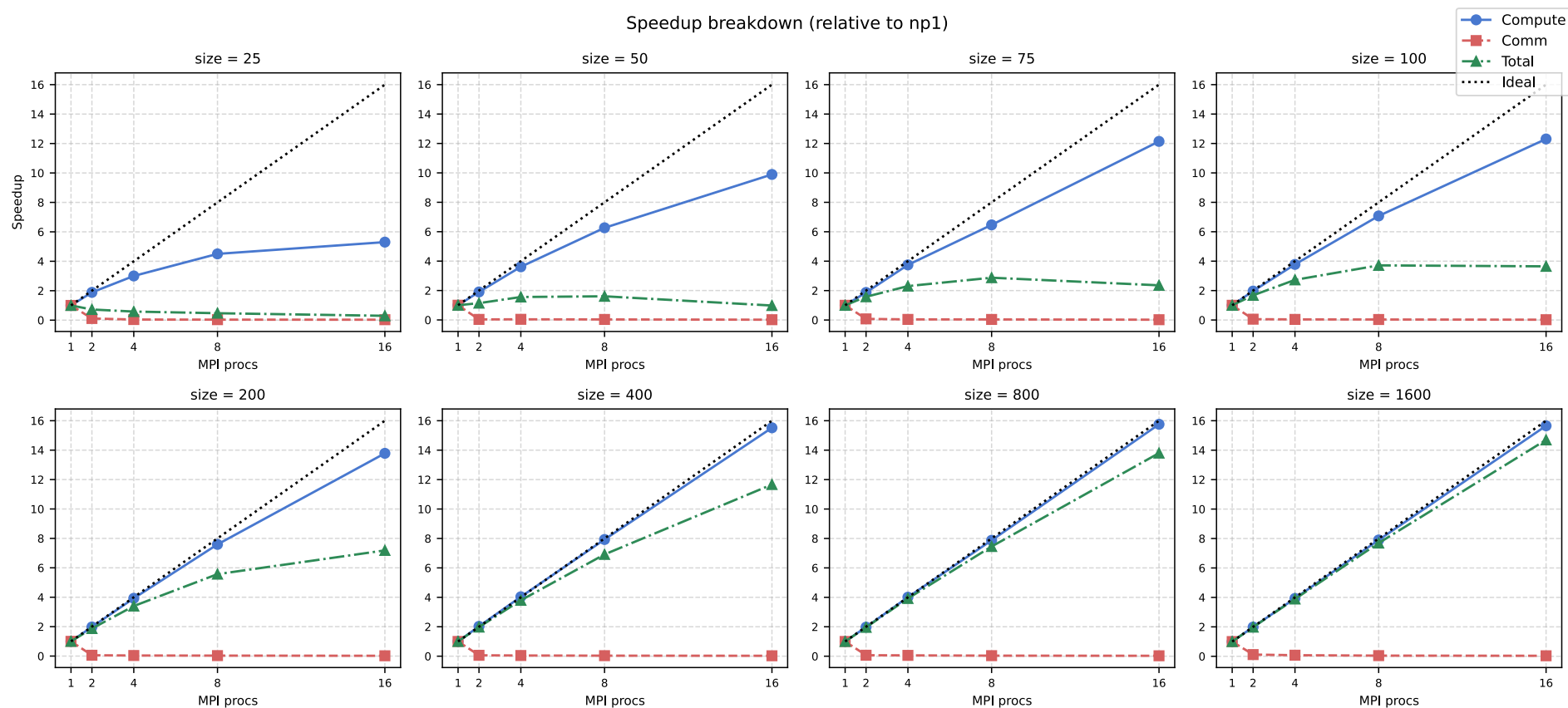


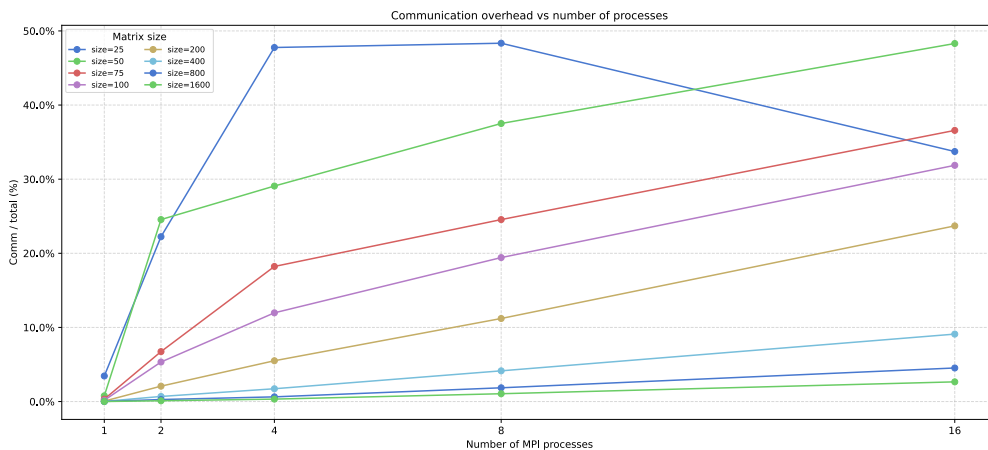
PART 3 : DISTRIBUTED MATRIX OPERATIONS (MPI)

Compute vs Communication time vs matrix size (log-log)



Speedup breakdown — compute / comm / total





- Pour les **petites matrices**, la communication domine
- Pour les **grandes matrices**, le compute redevient dominant
- Speedup proche de l'idéal pour les grandes tailles

Conclusion MPI

Bon speedup en compute, limité par l'Allreduce pour les petites tailles.

PART 4 : GPU MATRIX OPERATIONS (OPENCL)

- À compléter